TRUTHCHECK

Fact-checking health claims on YouTube.

An LLM agent that distinguishes settled medical consensus from genuine scientific debate — with citations.

Project review · May 2026

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OVERVIEW AND DATA COLLECTION

Why this project, and why now

THE PROBLEM

Claims made by YouTube videos might be misleading.

Example: 11% of the most-viewed YouTube videos on COVID-19 vaccines contradicted WHO/CDC guidance.

Li et al., BMJ Open, 2022

APPROACH

An agent, not a classifier.

- A claim like "seed oils cause inflammation" cannot be answered by a label.
- It is done by retrieving evidence, weighing study quality, and recognizing when scientists genuinely disagree.

AUTHORITATIVE SOURCES

- *YouTube video transcripts for Truthcheck .*
- *TruthCheck queries authoritative sources at inference, there is no training corpus (1 video at a time).*
- *The LLM provides reasoning; the sources provide ground truth.*

Queried live, per video

PubMed
peer-reviewed studies, via E-utilities API

Cochrane (through PubMed)
systematic reviews and meta-analyses

CDC / WHO / NIH
official guidance and consensus statements

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EXPLORATORY ANALYSIS

What does a real video look like?

SAMPLE CLAIM	VERDICT	WHY IT'S HARD
"Vitamin C deficiency causes scurvy."	Consensus supported	Universally agreed across centuries of evidence.
"Seed oils cause inflammation."	Contested	Real disagreement in current peer-reviewed literature.
"I feel better when I avoid gluten."	Out of scope	Personal anecdote, not a falsifiable claim.
"Vaccines cause autism."	Consensus contradicted	Conclusively refuted by large-scale studies.

> **Key insight:** The extractor's hardest job is filtering, not extracting.

EXPLORATORY ANALYSIS

2 Six pilot transcripts: clean, on-topic, but uneven

A pilot pass over six videos from two channels (Huberman Lab, Nutrition Made Simple). The corpus is well within model context limits and densely on-topic — the only real signal of trouble is caption punctuation, which falls off sharply for auto-generated transcripts.

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ON-TOPIC AND LOW-NOISE

All transcripts pass the 1% health-keyword density and <5% noise thresholds.

≤22%

OF THE 128 K CONTEXT WINDOW

Largest video (Rational Approach to Supplementation) uses ~29 k tokens. No chunking required.

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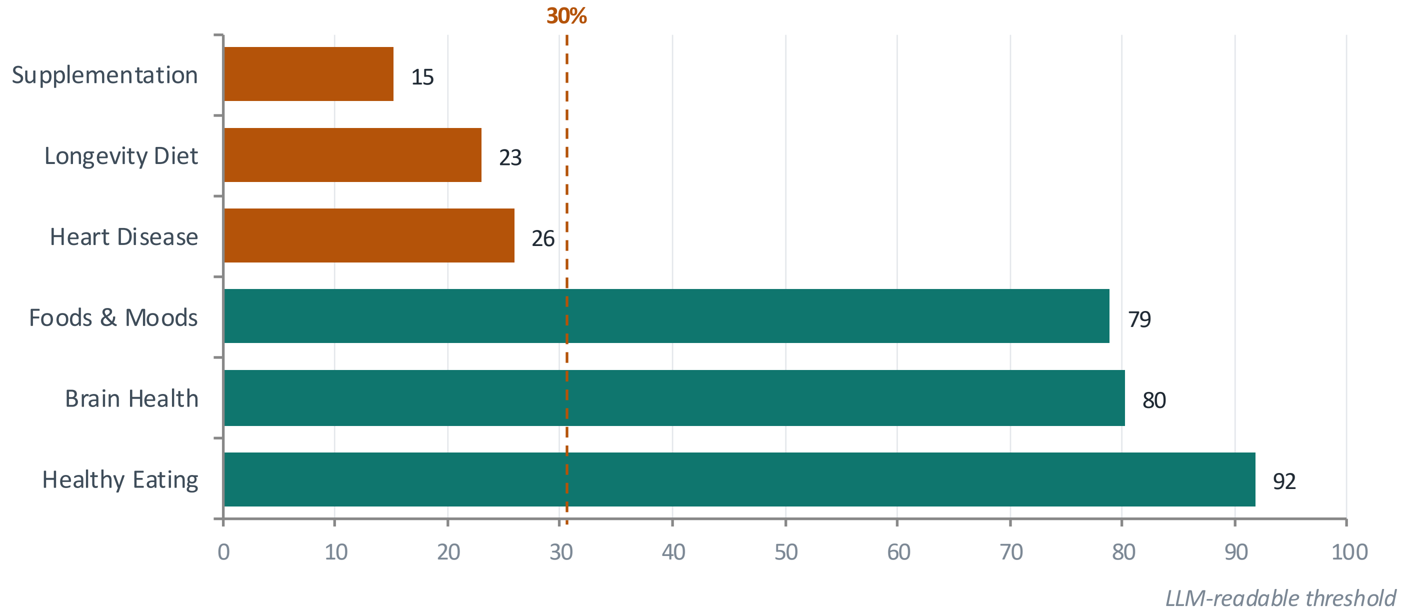
POORLY PUNCTUATED CAPTIONS

Auto-generated tracks fall to 15–26% punctuated segments. Restoration step needed pre-extraction.

PUNCTUATION RATE BY VIDEO

Manual captions vs. auto-generated



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MODELING APPROACH

A multi-stage agent pipeline

Five stages, each a focused LLM call or a fan-out across sources.
Pydantic schemas form the contract between.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Transcript	**Claim extraction**	**Evidence retrieval**	**Consensus eval**	**Report**
youtube-transcript-api				JSON + Markdown

ENGINEERING DECISIONS

- ✓ **Async I/O throughout**
- ✓ **LLM: Gemini-first, model-agnostic prompts.**
- ✓ **CI/CD: Git + GitHub Actions**
- ✓ **Deployment: managed cloud**

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TIMELINE

Milestones, status, and what could go wrong

- ✓ **Foundation** — Schemas, transcript, fixtures
- ✓ **Sources** — PubMed · Cochrane · CDC
- ◉ **Agents** — Extractor → retriever → eval
- ○ **Evaluation** — Run on 15-video set
- ○ **Polish** — CLI, formatter, demo video

● Done ◉ In progress ○ Pending

RISKS AND OPEN QUESTIONS

Evaluation methodology

No ground-truth dataset of labeled health claims at scale. Manual evaluation is the bottleneck.

Scope creep

Tempting to expand to general fact-checking. Resisting it is a design discipline, not a feature gap.

Thank you.

Questions and suggestions all welcome.

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ARCHITECTURE

Implementation notes for engineers

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